

(KTXFI2EBNF) Physics II. Exam 4 Solutions
February 2, 2026

I. Moving Reference Frames. Special Theory of Relativity

1. State the Galilean principle of relativity. (5 points)
Lecture on October 13, 2025, Slides 8–16.
2. Write down Einstein's relation expressing mass–energy equivalence. What is its physical meaning? (5 points)
Lecture on October 13, 2025, Slides 23–24.

II. Limits of the Classical Conceptual Framework

3. What is the photoelectric effect? Explain the mechanism of the photoelectric effect. (5 points)
Lecture on October 6, 2025, Slides 23–27.
4. Describe the Compton experiment and provide a qualitative but precise explanation of the phenomenon of Compton scattering. (5 points)
Lecture on October 6, 2025, Slides 29–31.
5. Formulate the de Broglie principle in your own words. (5 points)
Lecture on October 20, 2025, Slide 23.

III. Elements of Quantum Mechanics. Physics of Condensed Matter. Theory of Electrical Conduction

6. What is the tunneling effect? Describe the phenomenon in your own words. (5 points)
Lecture on November 10, 2025, Slide 46.
7. What is a forbidden band (band gap)? Define the concept. (5 points)
Lecture on October 27, 2025, Slides 18–19, 24–28.

IV. Development of Atomic Theory

8. Describe the Rutherford scattering experiment. (5 points)
Lecture on October 20, 2025, Slides 5–7.
9. What is the Zeeman effect? Describe the phenomenon. Under what conditions does it occur and how does it manifest itself? (5 points)
Lecture on October 20, 2025, Slides 16–17.
10. Given the element with atomic number 11 in the periodic table, write down its electronic configuration using the standard atomic physics notation. (5 points)
Lecture on October 20, 2025, Slides 19–22.

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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